
Behavioral Segmentation for Targeted Marketing

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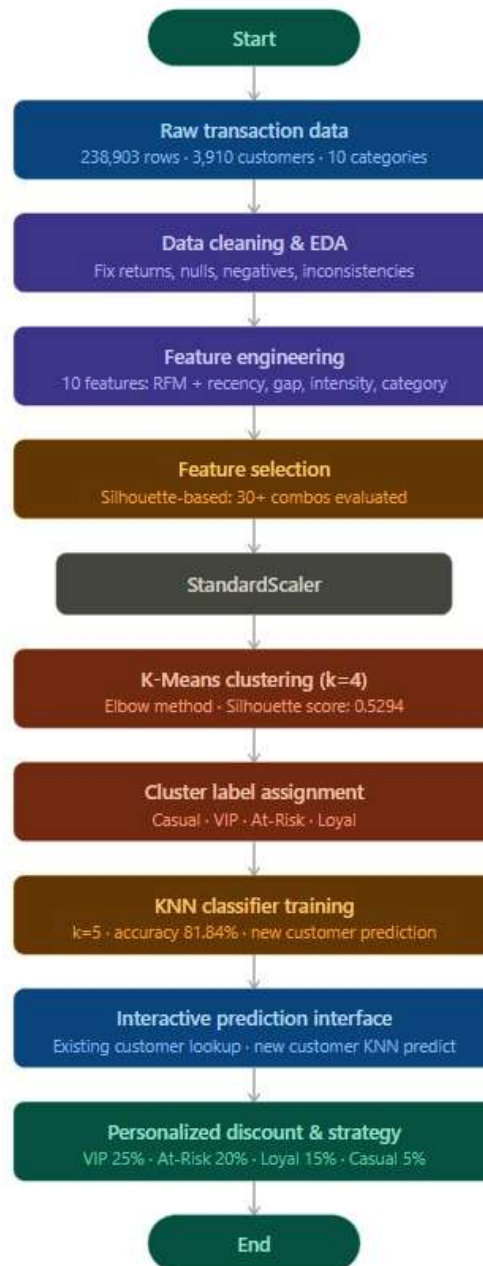
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1.Problem Description

The Behavioral Segmentation for Targeted Marketing system is a Python-based machine learning application developed to automate the process of grouping retail customers into distinct behavioral personas and generating personalized marketing strategies for each group. It streamlines the process of collecting transaction-level customer data, engineering behavioral features, training clustering and classification models, and producing actionable discount recommendations — all while maintaining high accuracy and interpretability. Using core concepts of Machine Learning such as data preprocessing, feature engineering, unsupervised clustering (K-Means and Hierarchical), supervised classification (KNN), model evaluation through Silhouette Score, and 2D/3D visualization via PCA, this project minimizes the manual effort required for customer profiling and supports data-driven targeted marketing decisions for retail businesses.

The system goes beyond the base project requirement of K-Means + Elbow visualization by introducing an original innovation: a trained KNN classifier that can predict the customer segment of any new customer in real time, based on just three input values. This makes the pipeline production-ready and applicable beyond the training dataset.

2. Architecture



3.Models Used(Applied/ Planned)

1.Logistic Regression

Introduction

Logistic Regression is a supervised machine learning algorithm mainly used for classification problems. In this project, Logistic Regression is used to predict whether a student's academic performance will be good, average, or poor based on factors such as attendance, internal marks, assignment scores, study hours, and previous academic records.

Working Principle

The model works by finding the probability of a particular outcome using a logistic function. It analyzes the relationship between dependent and independent variables and classifies students into different performance categories.

Why It Was Used

- Simple and efficient model for educational datasets.
- Provides fast prediction results.
- Easy to interpret and explain.
- Suitable for binary and multi-class classification.

Advantages

- Requires less computational power.
- Works well with structured data.
- Gives probability-based predictions.

Limitations

- Performance may decrease with highly complex data.
- Sensitive to outliers and irrelevant features.

2. Random Forest Classifier

Introduction

Random Forest is an ensemble machine learning algorithm that combines multiple decision trees to improve prediction accuracy. It is one of the most powerful supervised learning algorithms used in the project for predicting student performance.

Working Principle

The model creates several decision trees using different subsets of data and combines their outputs to generate the final prediction. This reduces overfitting and improves accuracy.

Why It Was Used

- Provides high prediction accuracy.
- Handles large datasets effectively.
- Works well with both numerical and categorical data.
- Reduces overfitting problems found in individual decision trees.

Advantages

- High accuracy and reliability.
- Handles missing values effectively.

- Provides feature importance analysis.

Limitations

- Requires more memory and computation.
- Model interpretation becomes difficult compared to simpler algorithms.

3.K-Means Clustering (Planned / Applied)

Introduction

K-Means Clustering is an unsupervised learning algorithm used to divide data into groups or clusters. In this project, K-Means clustering is planned and partially applied to group students according to their academic performance patterns.

Working Principle

The algorithm divides the dataset into K clusters based on similarity. Students with similar performance characteristics are grouped together.

Purpose in the Project

- To identify groups of high-performing and low-performing students.
- To analyze student behavior patterns.
- To provide targeted recommendations for different student groups.

Advantages

- Simple and fast clustering algorithm.
- Helps in discovering hidden patterns in data.
- Useful for performance segmentation.

Limitations

- Requires pre-selection of cluster count.
- Sensitive to initial centroid selection.

4. Hierarchical Clustering

Introduction

Hierarchical Clustering is another unsupervised learning technique used for grouping data points based on similarity. Unlike K-Means, it creates a hierarchy or tree-like structure of clusters.

Working Principle

The algorithm continuously merges similar data points or clusters until all points belong to a single hierarchy. The result is represented using a dendrogram.

Purpose in the Project

- To visualize relationships among students.
- To understand similarities in academic performance.
- To support detailed student segmentation.

Advantages

- No need to predefine the number of clusters.
- Provides clear visualization through dendrograms.
- Useful for detailed analysis.

Limitations

- Computationally expensive for large datasets.

- Difficult to handle very large amounts of data.

5. Feature Selection Techniques

Introduction

Feature Selection techniques are used to identify the most important attributes affecting student performance. This improves model efficiency and prediction accuracy.

Purpose in the Project

- To remove unnecessary features.
- To improve model performance.
- To reduce training time.
- To avoid overfitting.

Common Features Used

- Attendance Percentage
- Internal Marks
- Assignment Scores
- Study Hours
- Previous Semester Results
- Practical Performance

4.Evaluation metrics used

Introduction

Evaluation metrics are used to measure the performance and effectiveness of machine learning models. These metrics help determine how accurately the model predicts student outcomes and how reliable the system is for educational analysis.

Different metrics are used depending on whether the task is classification or clustering.

A.Classification Evaluation Metrics

1. Accuracy

Definition

Accuracy measures the percentage of correct predictions made by the model.

Formula

$\text{Accuracy} = (\text{Correct Predictions} / \text{Total Predictions}) \times 100$

Importance in the Project

Accuracy helps determine how effectively the system predicts student performance categories.

Advantages

- Simple to understand.
- Useful when dataset classes are balanced.

Limitation

- May give misleading results for imbalanced datasets.

2. Precision

Definition

Precision measures how many predicted positive results are actually correct.

Formula

$\text{Precision} = \text{TP} / (\text{TP} + \text{FP})$

Where:

- TP = True Positive
- FP = False Positive

Importance in the Project

Precision is useful when identifying students at risk of poor performance because false warnings should be minimized.

Advantages

- Reduces false positive predictions.
- Important for reliable recommendations.

3. Recall

Definition

Recall measures how many actual positive cases were correctly identified.

Formula

$\text{Recall} = \text{TP} / (\text{TP} + \text{FN})$

Where:

- FN = False Negative

Importance in the Project

Recall is important to ensure weak students are correctly identified by the system.

Advantages

- Helps identify maximum at-risk students.
- Improves educational intervention.

4. F1-Score

Definition

F1-Score is the harmonic mean of Precision and Recall.

Formula

$$\text{F1-Score} = 2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$$

Importance in the Project

It provides a balanced measure when both Precision and Recall are important.

Advantages

- Useful for imbalanced datasets.
- Gives balanced model evaluation.

5. Confusion Matrix

Definition

A Confusion Matrix is a table used to compare actual and predicted values.

Components

- True Positive (TP)
- True Negative (TN)
- False Positive (FP)
- False Negative (FN)

Importance in the Project

It helps analyze prediction errors and understand model performance in detail.

B. Clustering Evaluation Metrics

1. Silhouette Score

Definition

Silhouette Score measures how similar an object is to its own cluster compared to other clusters.

Range

The score ranges from -1 to +1.

- Close to +1 → Better clustering
- Close to 0 → Overlapping clusters
- Negative values → Incorrect clustering

Importance in the Project

Used to evaluate the quality of student grouping.

2. Inertia

Definition

Inertia measures the sum of squared distances between data points and their cluster centroids.

Importance in the Project

Lower inertia indicates better cluster formation.

Use

Used in the Elbow Method to determine the optimal number of clusters.

3. Dendrogram Analysis

Definition

A dendrogram is a tree-like diagram used in hierarchical clustering.

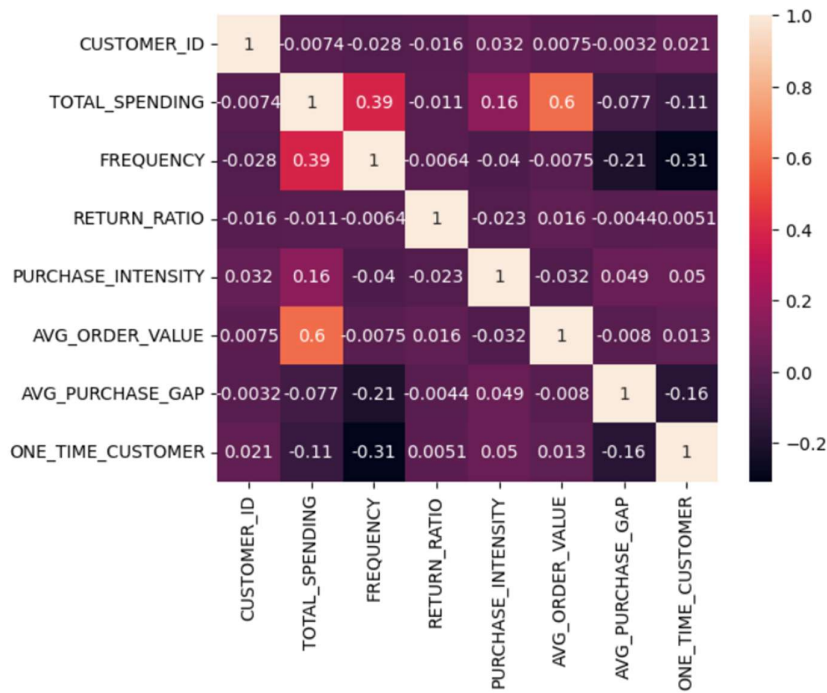
Importance in the Project

It helps visualize student similarity and cluster relationships.

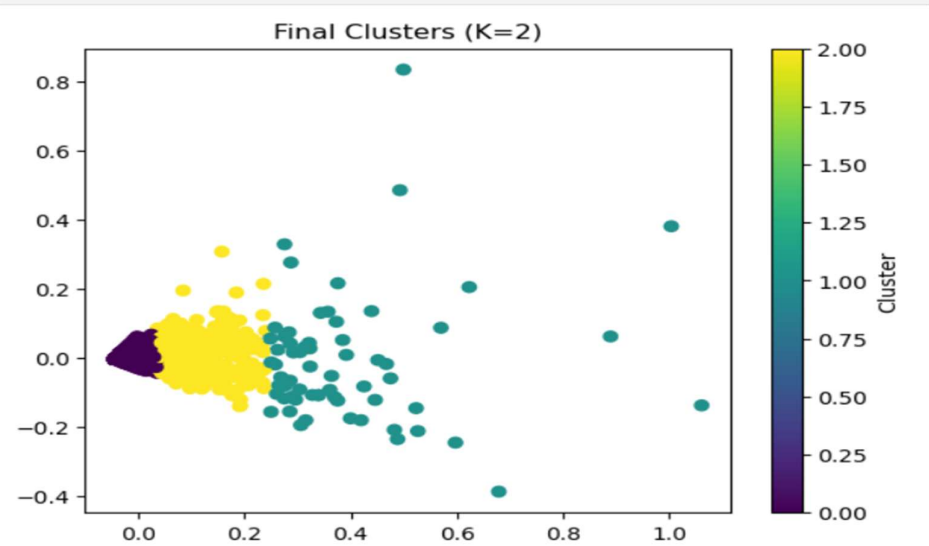
5. Preliminary Results and Visualizations

Heat Map

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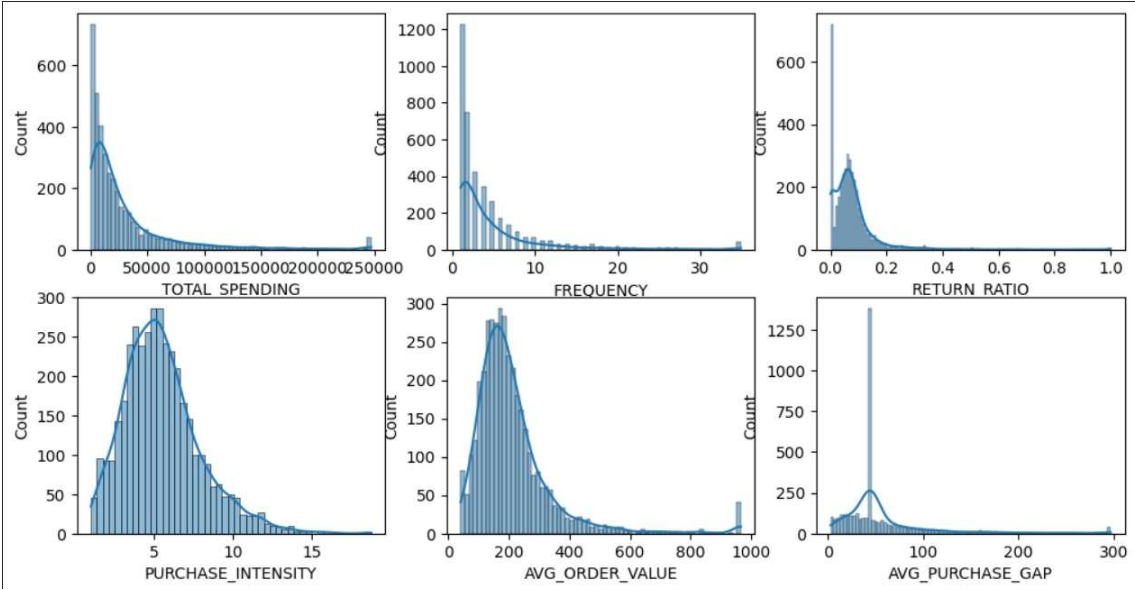


Cluster profile



Score

K=2, Score=0.3843921878248185
K=3, Score=0.38095719830572583
K=4, Score=0.3512582817539675
K=5, Score=0.3268583864746601
K=6, Score=0.3175447272253372
K=7, Score=0.3170342096710046
K=8, Score=0.29138953311497867
K=9, Score=0.3046599749055303



Silhouette Score: 0.24411841273336712

6. References:

Books:

1. Let Us Python – Yashavant Kanetkar

Online Resources:

- GeeksforGeeks – <https://www.geeksforgeeks.org/>
- TutorialsPoint – <https://www.tutorialspoint.com/cprogramming/index.htm>
- Stack Overflow – <https://stackoverflow.com>